

RAVINDRA SINGH

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RESEARCH PROFILE

Delhi is officially one of India's greenest cities. It is also a city where millions of residents in informal and unauthorised settlements live without meaningful access to green space, cooling shade, or clean urban environments. This paradox is the starting point of my doctoral research. Spatial justice is the organising framework, asking not just whether environmental resources are distributed fairly (distributive justice), but who participates in decisions about urban environments (procedural justice), and whose communities and claims are recognised as legitimate within planning systems (recognition justice). My central argument is that misrecognition is foundational: when planning frameworks refuse to acknowledge informal settlements as legitimate urban communities, procedural exclusion and distributive deprivation follow structurally. My work goes beyond green cover counts and heat metrics. Through geospatial modelling, critical master plan analysis (1962 to 2041), and governance fieldwork in urban villages and unauthorised colonies, I examine how Delhi's greening agenda concentrates ecological benefits in planned areas while displacing environmental costs onto the urban poor.

EDUCATION

PhD in Geography

2021 – Present

Centre for the Study of Regional Development (CSRD), Jawaharlal Nehru University, New Delhi

Dissertation: Socio-economic Embeddedness of Urban Green Spaces in Delhi: A Spatial Justice Framework

Supervisor: Professor Milap Punia

M.A. in Geography

2016 – 2018

Central University of Punjab, India

Specialisation: Urban and Environmental Geography; GIS and Remote Sensing

Dissertation: Mapping of Industrial Waste of Granite and Marble Slurry in Rajasthan

RESEARCH AND PROFESSIONAL EXPERIENCE

Consultant — Shahjahanpur City Heat Action Plan 2026

2025 – 2026

District Disaster Management Authority, Shahjahanpur, Uttar Pradesh

Commissioned to lead spatial heat risk assessment and vulnerability mapping underpinning the full municipal Climate Adaptation Plan. Produced Chapter 2 (Geospatial Heat Risk Assessment), developing a composite Heat Risk Index using Geospatial data. Provided substantive policy inputs across Guiding Principles, Early Warning Systems, Operational Heat Action Strategy, and department-wise Standard Operating Procedures, translating spatial evidence into actionable governance frameworks. One of the few instances of doctoral research directly shaping a published city-level policy instrument in India.

[Published plan: shahjahanpur.nic.in/dmp/](https://shahjahanpur.nic.in/dmp/)

Senior Research Fellow — UGC Fellowship

2021 – Present

Centre for the Study of Regional Development, Jawaharlal Nehru University

- Designing and conducting mixed-method research integrating geospatial modelling, critical master plan analysis (1962 to 2041), and field-based governance inquiry.
- Running the InVEST Urban Cooling Model to quantify distribution of heat mitigation benefits across settlement typologies (Unauthorised colonies vs slum vs Urban Villages vs Rest of Delhi); key finding: Multiple times cooling deficit in Informal Settlements relative to planned settlements.
- Developing spatial justice indices through raster-based modelling, proximity analysis, weighted overlay techniques, spatial regression, and Gi* hotspot analysis to assess environmental inequality.

- Examining distributive, Governance, and Cultural dimensions of urban green infrastructure governance in Delhi using Roberto Rocco's spatial justice framework.
- Conducting community-level fieldwork in Delhi's urban villages in collaboration with independent researcher Puneet Singh Singhal (Dilli Dehat Project), examining how conservation boundaries and planning exclusions dispossess historically documented settlements.
- Producing policy-relevant spatial evidence to interrogate planning hierarchies and environmental allocation patterns.

Intern

2019

Indo Global Social Service Society (IGSSS), New Delhi

Contributed to the Urban Agenda 2041 initiative, including climate justice analysis and policy manifesto preparation addressing environmental inequality in Delhi.

Diploma

Urban Fellowship Award

2018 – 2019

Indian Institute for Human Settlements (IIHS), Bangalore

Nine-month fully residential, interdisciplinary fellowship at India's only nationally recognised urban practice training institution. Curriculum combined classroom instruction, site-based applied learning, and live project work across governance, ecology, infrastructure, and spatial planning. Developed foundational skills in translating complex urban research into practice-oriented governance interventions, working across the full spectrum of urban themes, including water, sanitation, commons, planning, and identity. IIHS draws faculty from India's leading urban institutions and international partners.

FELLOWSHIPS AND SCHOLARSHIPS

UFS Futuring Summer School — Futuring for Democracy

July 2026

Utrecht University, Netherlands

Competitive grant of EUR 2,000 (fee waiver, travel, accommodation, and meals) for an intensive programme on democratic futures, participatory scenario planning, and governance innovation for sustainability transitions.

Amsterdam Complexity School on Climate Change — Travel Grant

April – May 2026

UvA Institute for Advanced Study, Amsterdam, Netherlands

Competitive travel grant (EUR 800) plus full accommodation and meals. One of the most selective cohorts to date, with applications from almost every country and numbers doubling from the previous year.

Austrian Student Union Travel Grant

2025

Vienna Doctoral School, Austria

Competitive travel grant (EUR 800) for participation in the workshop on Urban Space, Financialisation, and the Right to the City.

Indo-German Centre for Sustainability Fellowship — Regenerative Urban Futures

2024

IGCS, RWTH Aachen, Germany

Fully funded fellowship (EUR 2,000) covering travel, fees, and accommodation for the programme on Circular Economies and Societies.

TU Delft Summer School Scholarship — Planning and Design for the Just City

2024

Centre for the Just City, Faculty of Architecture and Built Environment, TU Delft, Netherlands

Competitive scholarship (EUR 500) plus fees and accommodation for the 10th anniversary edition of this two-week intensive school on spatial justice, climate adaptation, and urban sustainability. One of a limited number of scholarships awarded globally.

University Grants Commission Junior Research Fellowship

2021 – Present

Government of India

Nationally competitive fellowship awarded to top doctoral candidates across India, funding full PhD research at JNU. Selection based on national-level NET examination across all disciplines.

PUBLICATIONS AND POLICY REPORTS

Jasrotia, G., Agarwal, O., Singh, R., Choudhary, R. S., and Athar, T. (Forthcoming 2026). A Manifesto for a Waste-Just Delhi: How Delhi Produces Waste Injustice and How We Must End It. In R. Rocco et al. (Eds.), A Manifesto for the Just City (Vol. 5). TU Delft Open Publishing.

Singh, R. (2026). Geospatial Heat Risk Assessment and Spatial Analysis [Chapter 2, Technical Expert]. In Shahjahanpur City Heat Action Plan: A Heatwave Management Guide for 2026. District Disaster Management Authority, Shahjahanpur, Uttar Pradesh.

Jasrotia, G., Punia, M., Singh, K. K., and Singh, R. (2025). A manifesto for environmental justice in Delhi. In R. Rocco, C. Newton, and J. Goncalves (Eds.), A Manifesto for the Just City (Vol. 4, pp. 380-389). TU Delft Open Publishing. <https://doi.org/10.59490/mg.142>

CONFERENCE PRESENTATIONS

LivedHeat Symposium 2026 May 2026
IIT Delhi and TU Delft, New Delhi

Who Gets to Stay Cool? Spatial Inequality in Urban Heat Mitigation across Delhi.

Spatial Planning in Practice International Conference 2023
TU Delft, Netherlands

Assessment of Socio-economic Embeddedness of Urban Green Spaces in Delhi.

43rd INCA International Congress 2023
Indian National Cartographic Association, hosted by NRSC, ISRO

Geospatial Assessment of Quality, Quantity and Accessibility of Urban Green Spaces in Delhi.

TRAINING AND WORKSHOPS

School of Climate Change (SoCC) Certificate Course Jan – Mar 2026
Oxford Climate Society

Eleventh Annual PhD Workshop 2023
Indian Institute for Human Settlements (IIHS), Bangalore
Sessions on academic writing, non-academic writing, and research communication.

NGP-DST Winter School on Advances in Geospatial Techniques Dec 2021
Institute of Environment Education and Research, Bharati Vidyapeeth University, Pune
21-day Level 2 programme on geospatial techniques in ecosystem conservation and management. Mini Project: Forest Landscape Analysis for Uttarakhand, 2005 to 2020.

Summer Training Programme in Remote Sensing, GIS, and GPS May – Jun 2018
Punjab Remote Sensing Centre (PRSC), Ludhiana
Six-week programme on cadastral mapping, remote sensing, GIS, and GPS.

Course: Exploring Sustainability in the Indian Context Apr – Jul 2022
Azim Premji University, India

SERVICE AND PROFESSIONAL RESPONSIBILITIES

Rapporteur Jan 2022
UN-GGIM Pre-event, Centre for the Study of Regional Development, JNU, New Delhi

Rapporteur for multiple sessions at the United Nations Global Geospatial Information Management pre-event.

Rapporteur

Aug 2023

International Conference on Sustainable and Inclusive Urban Development in the Global South, New Delhi

Organised by IHD, NITI Aayog, University of Florida, CSRD-JNU, and National Institute of Urban Affairs.

Student Representative, Board of Studies

2017 – 2018

Centre for Geography and Geology, School of Environment and Earth Science, Central University of Punjab

METHODOLOGICAL EXPERTISE

My research integrates three methodological registers: geospatial modelling and remote sensing (ArcGIS Pro, QGIS, Google Earth Engine, InVEST) to quantify how environmental benefits are distributed across settlement typologies; spatial statistics including regression, hotspot analysis, and Lorenz curves to identify structural patterns of inequality; and governance fieldwork and master plan analysis to trace how planning decisions produce exclusion on the ground. Numbers identify the pattern. Fieldwork explains the mechanism.

Research Methods: Mixed-method research design, critical master plan analysis, qualitative governance fieldwork, community interviews, participatory mapping, institutional analysis, policy document analysis.

LANGUAGES

Rajasthani (mother tongue) · Hindi (excellent) · English (excellent)